Describe Data by Range and Interquartile Range

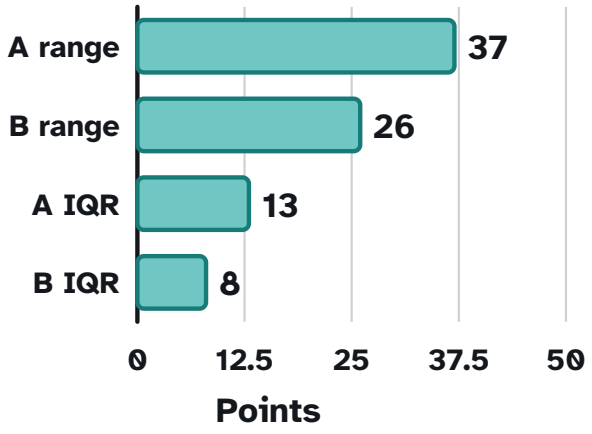
Use the range and the IQR together to say which group is more consistent.

Name _____ **Date** _____

Tonight's spark

Two classes took the same quiz. Class A scores run from 52 to 89. Class B scores run from 57 to 83. Which class has scores closer together?

Make a guess first: Class A or Class B?



Our class model: range and IQR side by side

Try it

Work it out on paper. Write one answer on each line.

1 **A smaller range or IQR means the data values are...**

A. Closer together, so more consistent **B.** Farther apart, so more consistent
C. Greater in value **D.** Always equal

My answer _____

2 **Class A has a range of 20 and Class B has a range of 12. Which class is more consistent?**

A. Class A **B.** Class B **C.** They are equally consistent **D.** You cannot tell

My answer _____

3 **Team R has an IQR of 6 and Team S has an IQR of 10. Whose middle half is more spread out?**

A. Team R **B.** Both are the same **C.** Team S

D. Neither, because IQR is not about spread

My answer _____

4 **A class has least 50, Q1 58, median 65, Q3 70, greatest 90. Which gives the range and the IQR?**

A. Range 40, IQR 70 **B.** Range 90, IQR 12 **C.** Range 12, IQR 40 **D.** Range 40, IQR 12

My answer _____

5 **Group X has range 30 and IQR 10. Group Y has range 25 and IQR 14. Which sentence is true?**

A. X has the larger range, and Y has the larger IQR **B.** X is larger in both

C. Y is larger in both **D.** They are the same in both

My answer _____

Talk about it

Do the range and the IQR point to the same team? What does that tell you?

Home mission

Think of two ways to get to school or the store. Which one always takes about the same time? Which one is sometimes fast and sometimes slow? Which has the smaller spread?

If we get stuck, ask

- Is this measure a subtraction? Which two numbers are we subtracting?
- Could the middle half ever be wider than the whole set?
- Which team's numbers sit closer together?

Do it on a screen

Scan or type the link. Same lesson, with hints.

eduwonderlab.com/lessons/2-5-part2/homework.html

How did it feel? (optional)

Tricky Getting there Got it